OCHMO Office of Research Assurance

Human Subjects Research Guidance

Guidance - 415: Human Subjects Research Determination

Number	Effective Date	Version Number	Previous Versions
415	2/4/2025	v.3	v.2

Purpose:

The information in this guidance is intended to aid Principal Investigators (PI) in determining whether certain types of projects, such as human in the loop (HITL) studies, quality improvements activities, engineering studies, or use of data/specimens are subject to oversight by the NASA Institutional Review Board (IRB).

Activities meeting the regulatory definitions for research with human subjects (Common Rule) or a clinical investigation (FDA) must have prospective IRB review and approval or determination of exemption. Activities that do not meet the regulatory definitions are considered “Not Human Subjects Research” (NHSR) and are not subject to oversight by the IRB.

The guidance in this document focuses on activities meeting the Common Rule (i.e., common set of human subjects regulations agreed on by 20 US Federal Agencies) definitions for *research* with *human subjects*. NASA is an official signatory of the Common Rule, 45 CFR part 46, codified at 14 CFR part 1230. The Food and Drug Administration (FDA) has a separate set of regulations that are beyond the scope of this document. PI conducting activities with one or more human subjects in which the results are intended to be submitted to the FDA should consult with NASA IRB well in advance.

NASA SOP-405 outlines NASA policy and the administrative process for obtaining documentation of NHSR through the NASA IRB. PI make an initial assessment whether a project is or is not Human Subjects Research (HSR) as part of their submission process. Making this assessment can be nuanced and context specific. While the aim of this document is to outline criteria for making these initial assessments, PI are encouraged to consult with NASA IRB staff whenever there is uncertainty.

In this document:

Who Decides
What does the IRB oversee
  Is it "research"?
  Does the research involve human subjects?
Examples
Is IRB Approval Required?
Definitions
Additional Information

Who decides?

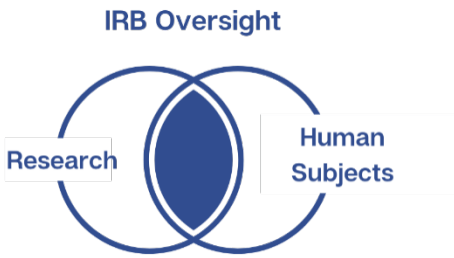
Only the NASA IRB Staff, the NASA IRB Chair, or the convened NASA IRB are authorized to make an ***official IRB determination*** that an activity does or does not involve human subjects research (i.e., Not Human Subjects Research (NHSR)). While the federal regulations do not mandate that researchers obtain official NHSR documentation from the IRB, other NASA review boards (e.g., TRR, Safety) may require formal documentation. PI may request EDRO documentation, rather than an official IRB determination, when the activities clearly fall outside of the federal definitions of human subjects research and do not involve International Partner (IP) research approved by an international ethics board.

PI may make an initial assessment that an activity is Not Human Subjects Research (NHSR) using this guidance, the NHSR Questionnaire, and/or through consultation with NASA IRB staff. The process for requesting a formal determination is outlined in HRP-405.

- The NASA IRB's determination that an activity is human subjects research supersedes the determination of another IRB when NASA is engaged.
- The NASA IRB may make an NHSR determination based on information provided in a full-board/expedited/exempt submission.
- The NASA IRB does not have authority to grant retroactive approval for a project meeting the definitions of research with human subjects.

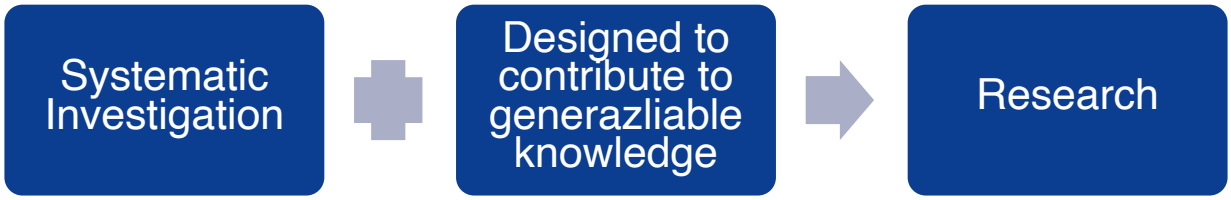
What does the IRB Oversee?

Projects subject to IRB oversight are those that meet the federal definitions for **both** "*research"* and "*human subjects*." The first consideration is whether the proposed project is "***research.***"



Is it “research”?

Human in the loop (HITL) studies, quality improvements activities, engineering studies quality assessment/ improvement initiatives and other similar projects frequently involve interactions with, and the collection of information from, people in ways that may or may not be considered to be *research* with human subjects. A project must involve both a *systematic investigation* and be *designed to contribute to generalizable knowledge* to meet the definition of “*research*.”



Systematic investigation means that a prospective plan is followed to gather information. A *systematic investigation* is not limited to quantitative data collection and includes qualitative methodologies. While the methods and organization of qualitative and quantitative inquiry look different, for both, the researcher formulates a plan that includes (1) ideas about what they would like to learn and (2) how they intend to learn it.

Activities that contribute to *generalizable knowledge* are those that:

- are those designed to draw broadly applicable conclusions, or
- generalize findings beyond a single individual or internal program, or
- uncover underlying principles that have predictive value and can be applied to other circumstances, or
- develop or test scientific theories or hypotheses.

NASA frequently engages in activities that require the efforts of and/or interaction with humans. These projects typically involve *systematic investigations*; however, often the purpose is narrow and limited to a specific context for which outcomes are not intended to have broad applicability or advance overall knowledge within a field.

For example, verification that equipment or systems can withstand X number of hours of human use prior to deployment to ISS involves a systematic investigation and humans, but the project is not designed to contribute to generalizable knowledge. Accordingly, IRB oversight would not be required, provided the intent and scope of the activities remain limited.

The intent to publish or present findings *suggests* that a project may be *designed to contribute to generalizable knowledge*; however, publication alone does not define a project as *"research.* Non-research endeavors are frequently disseminated for administrative, documenting, reporting purposes, or to share with interested audiences (e.g., as case reports, lessons-learned, etc.) without an intent to generalize outcomes to other situations, programs, or populations.

Publishing a case-report of a specific scenario that allows readers to draw their own conclusions of applicability (i.e., not designed to contribute to generalizable knowledge) does not constitute research simply because it published. Conversely, activities that **are designed** to draw broadly applicable conclusions, identify themes that can be applied broadly, etc. are considered *research*, even if not published.

PUBLICATION

The intent to publish is an insufficient criterion for determining whether an activity involves research.

Planning to publish an account of a quality improvement project does not necessarily mean that the project fits the definition of research; people seek to publish descriptions of non-research activities for a variety of reasons, if they believe others may be interested in learning about those activities.

Conversely, a project may involve research even if there is no intent to publish the results.

[OHRP Quality Improvement Activities-FAQs]

An activity must be both a *systematic investigation* and *designed to contribute to generalizable knowledge* to meet the definition for *research.* Failure to meet the definition for one or the other means that it does not constitute *research* and IRB oversight is not necessary – regardless of whether an activity involves *human subjects.*

Does it involve "Human Subjects"

If a project has been determined to meet the regulatory definition of *Research*, activities should be assessed regarding the involvement of *Human Subjects.*

The regulations define a *Human Subject* as a *living individual about whom* and investigator conducting *research*:

> (i) obtains information or biospecimens through *intervention or interaction* with the individual and uses, studies, or analyzes the information or biospecimens;
>
> (ii) obtains, uses, studies, analyzes, or generates *identifiable private information* or biospecimens.

This definition encompasses *research* that actively involves interaction/intervention with individuals and/or the secondary use of data or specimens.

LIVING INDIVIDUAL

IRB only oversees research with living participants. Research conducted with cadavers, biospecimens from, or data about deceased individuals does not fall under IRB oversight.

ABOUT WHOM

Research information is "about" an individual when it involves their behaviors, attitudes, opinions, personal characteristics, demographics, decisions, reactions to stimuli, physical or physiological responses, biological specimens, etc. Data or information that involves *only* factual, objective information on something other than the individual (e.g., factual information about an organization, procedure, or product) is not about the individual and thus not considered to involve *human subjects*. However, if questions include individuals' opinions or attitudes that information becomes about the individual and therefore the project is considered to involve human subjects.

ABOUT WHOM…

Information involving an individual's:

- Behaviors
- Opinions
- Beliefs
- Attitudes
- Responses to stimuli
- Physical or physiological responses
- Personal characteristics
- Demographics
- Biological specimens

Thus, the determination of whether a project does or does not involve *human subjects* can hinge on question wording. For example, one would expect individual reports of factual information to not vary depending on who responds (e.g., How many administrative employees does company X employ?); whereas subjective information *about* an individual's opinions/attitudes/beliefs varies depending on the individual responding (e.g., Does company X employ an adequate number of administrative staff?). With the former, there is a factual answer regarding the number of employees that should be the same (or if not, someone is incorrect); with the latter, the opinions regarding the adequacy of administrative support levels are likely to vary depending on who you ask.

INTERVENTION OR INTERACTION

Prospective information about individuals can be obtained or generated through ***intervention*** or ***interaction*** with human subjects. Intervention includes both physical procedures by which information or biospecimens are gathered (e.g., venipuncture) and manipulations of the subjects' environment performed for research purposes. Interaction can include communication or interpersonal contact between the investigator and subject as well as collecting information through observation of behavior.

Examples:

- Physical or biomedical procedures (imaging, scanning, body measurements, collection of biospecimens)
- Use of physical sensors applied to the surface of the body
- Imaging (MRI, x-ray)
- Electrocardiography, electroencephalography, thermography, electroretinography, ultrasound, doppler blood flow, echocardiography
- Use of instruments, devices, apps, software (e.g., video, screen capture, eye-tracking, keylogging) to collect data or monitor behavior
- Surveys, interviews, questionnaires, focus groups
- Tasks or experiments in physical or electronic environments

SECONDARY USE OF DATA/BIOSPECIMENS

Projects can fall under the umbrella of IRB oversight even when they do not involve direct work with human participants. For example, some research with biological samples *may* be considered HSR, even if the samples were collected for a different study. The need for IRB oversight for this secondary use hinges on whether the information/specimens can be considered *identifiable private information* or biospecimens.

Private information includes information about behavior that occurs in a context in which an individual can reasonably expect that no observation or recording is taking place and information that has been provided for specific purposes that the individual can reasonably expect will not be made public (e.g., medical record).

Identifiable private information is private information for which the identity of the subject is, or readily may be ascertained by a member of the research team or associated with the information. Information can be considered *directly* or *indirectly identifiable. Directly identifiable* means that the identifying information (e.g., names, email address, facial images, phone numbers) are included with the information/specimens either embedded with or linked to the data and a member of the research team is able to re-identify the information. Data can be considered *indirectly identifiable* when obvious direct identifiers have been removed, but one may still be able to ascertain the identity of subjects through a *combination* of information contained in the data.

The NASA IRB considers secondary research to involve human subjects *unless* it meets both of the following conditions:

- The research team is NOT collecting information/specimens through interacting or intervening with living people; and
- None of the research team (i.e., investigators or collaborators/co-authors, personnel) can identify the subjects through coded private information or specimens at any point during the research and no attempts will be made to reidentify participants.

If any investigator involved in the research can determine a subject's identity or has access to identifiers, the research is considered to involve human subjects and IRB review requirements apply.

Do I need to submit an IRB application?

Research projects that involve *human subjects* require prospective IRB approval. PI are encouraged to assess whether a project meets the definition of research such that IRB oversight is required before submitting an application for review. PI's may submit an NHSR

Questionnaire through the eIRB if documentation that a project is "Not Human Subjects Research" (thus not requiring IRB oversight) is needed.

Individuals conducting program evaluations, quality assessment/improvement or other evaluative activities who are uncertain about whether IRB approval is required may:

- Complete the NHSR Questionnaire
- Consult the HRP-405
- Contact IRB staff for guidance on whether a specific project meets the definition of research.

Contact IRB staff via email nasa-irb@nasa.gov

Definitions

Applicable to all activities: 2018 Common Rule [14 CFR 1230.102, 45 CFR 46.102]

Generalizable Knowledge – Activities designed to develop or contribute to generalizable knowledge are those designed to draw broadly applicable conclusions or generalize findings beyond a single individual or internal program, uncover underlying principles that have predictive value and can be applied to other circumstances, or develop or test scientific theories or hypotheses.

Human Subject – a living individual about whom and investigator conducting research [14 CFR1230.102(e)(1)]

i) obtains information or biospecimens through intervention or interaction with the individual and uses, studies, or analyzes the information or biospecimens; or

ii) obtains, uses, studies, analyzes, or generates identifiable private information or biospecimens.

Identifiable private information – private information for which the identity of the subject is or may be readily ascertained (directly or indirectly), by the investigator (or others) or associated with the information. [14 CFR1230.102(e)(5)]

Interaction – communication or interpersonal contact between investigator and subject. [14 CFR1230.102(e)(3)]

Intervention – includes both physical procedures by which information or biospecimens are gathered and manipulations of the subject or the subject's environment that are performed for research purposes. [14 CFR1230.102(e)(2)]

Private information – information about behavior that occurs in a context in which an individual can reasonably expect that no observation or recording is taking place, as well as information that has been provided for specific purposes by an individual and that the individual can reasonably expect will not be made public. [14 CFR1230.102(e)(4)]

Research – a systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge. [14 CFR1230.102(l)]

Systematic investigation – Examination that involves a prospective plan that incorporates data collection, either quantitative or qualitative, and data analysis to answer a question.

In addition to the above definitions, activities regulated by the Food and Drug Administration (FDA) also apply the FDA definitions [21 CFR 50.3]

Clinical Investigation -Any experiment that involves a test article and one or more human subjects that is either subject to requirements for prior submission to the Food and Drug Administration under section 505(i), or 520(g) of the act, or is not subject to requirements for prior submission to the Food and Drug Administration under these sections of the act, but the results of which are intended to be submitted to, or held for inspection by the Food and Drug Administration as part of an application for a research or marketing permit." The term does not include experiments that are subject to the provision of 21 CFR 58, regarding nonclinical laboratory studies. [21 CFR 50.3(c)]

Note: Under FDA regulations, the terms "research" and "clinical investigation" are synonymous.

Test Article - A test article means any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other article subject to regulation under the Federal Food, Drug and Cosmetic Act [21 CFR 50.3(j)].

Human Subject (FDA) - an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient. [21 CFR 50.3(g)]

Additional Information:

Quality Improvement Activities FAQs
US Department of Health and Human Services I OHRP I Regulations & Policy IGuidance I FAQ

Version #	Version Notes:	Originated By:	Approved By:	Date Approved:
v.3	Minor revisions	J. Kisenwether	J. Kisenwether	27March2025
v.2	Minor revisions	A. Garza	M. Covington	04Feb2025
v.1	Original	J. Ensley Gorshe	M. Covington	19May2022